

Product-to-MCP User Fields Checklist

Shareable step-wise intake fields for creating, testing, deploying, and publishing an MCP server from a product API.

One-line flow

Project details -> API auth -> OpenAPI upload -> review operations -> create actions -> validate/approve actions -> create publishing profile -> generate MCP release -> test tools -> deploy public HTTPS backend -> publish to Smithery

Step 1: Product/API Setup

Field	Required	Notes
Project name	Yes	Example: Demo Store
API base URL	Yes	Stable API root, not one endpoint. Example: https://api.customer.com/v1
Authentication type	Yes	Bearer token, API key header, or No authentication
Upstream API key/token	If auth is not none	Stored server-side; never shown in MCP tool schema
Credential header	If auth is not none	Example: Authorization, x-api-key

Step 2: OpenAPI Import

Field	Required	Notes
OpenAPI file	Yes	OpenAPI 3.0/3.1 JSON or YAML
OpenAPI paste content	Optional	Backend supports file upload or raw content
API environment/version	Recommended	Example: production, staging, v1
Operations business meaning	Recommended	Needed when OpenAPI descriptions are unclear

System Generated Value	Notes
Operations	Methods, paths, params, schemas
Operation groups	Auto-grouped from tags/path
Supported/unsupported status	Unsupported operations are shown, not guessed

Step 3: Action Studio

The user selects and configures what the agent can actually use.

Field	Required	Notes
Operation group name	Optional	Can rename discovered groups
Hidden groups	Optional	Hide irrelevant operation groups

Field	Required	Notes
Selected operation(s)	Yes	One action can use one or more approved API operations
Tool/action name	Yes	Lowercase snake case, max 64 chars. Example: get_product
Action title	Yes	Human-readable name
Action description	Yes	What the tool does
Agent input schema	Yes	JSON Schema for what the agent can pass
Action data/output schema	Yes	JSON Schema for action result
Execution steps	Yes	Ordered API steps
Argument bindings	Yes	Maps agent input, previous step output, or constants to API params/body
Run condition	Optional	For multi-step actions
Normalized output fields	Optional	Select exact fields returned to agent
Test arguments JSON	Recommended	Used to validate the action before publishing

Required User Decision	Notes
Which operations become actions	OpenAPI alone does not grant permission
Whether writes are allowed	Prototype supports CRUD, but writes require explicit selection
Which fields are exposed/redacted	Especially for PII, finance, health, confidential data

Step 4: Publishing Profile

This decides which approved actions appear in the MCP.

Field	Required	Notes
Profile name	Yes	Example: Customer Support
Profile description	Yes	Example: Focused tools for support agents
Approved actions	Yes	Only approved actions can be added
Large profile confirmation	Conditional	Required if profile has more than 30 tools

Generated Value	Notes
Profile ID	Internal
Immutable MCP release	Created from the profile
Release ID	For support/debugging
Deployment slug	Used in MCP URL
Manifest hash	Identifies exact immutable release
Public MCP URL	Used by Smithery/MCP clients
Local MCP URL	Only for local testing

Step 5: MCP Testing

Field	Required	Notes
Tool test arguments	Recommended	JSON arguments per generated tool

System Shows	Notes
Tool list	What MCP clients will see
Tool response	Result from upstream API
Errors/warnings	Validation, timeout, upstream failure, etc.

Step 6: Deploy Backend/Frontend

Backend environment fields:

```
PRODUCT_TO_MCP_ENV=production
PRODUCT_TO_MCP_DATABASE_URL=postgresql://USER:PASSWORD@HOST:5432/DATABASE
PRODUCT_TO_MCP_PUBLIC_BASE_URL=https://api.your-domain.com
PRODUCT_TO_MCP_CORS_ORIGINS=https://app.your-domain.com
PRODUCT_TO_MCP_ALLOWED_HOSTS=api.your-domain.com
PRODUCT_TO_MCP_SMITHERY_API_URL=https://api.smithery.ai
PRODUCT_TO_MCP_MAX_OPENAPI_BYTES=20971520
PRODUCT_TO_MCP_MAX_CHAIN_STEPS=10
PRODUCT_TO_MCP_ACTION_TIMEOUT_SECONDS=60
PRODUCT_TO_MCP_UPSTREAM_TIMEOUT_SECONDS=20
PRODUCT_TO_MCP_MAX_UPSTREAM_RESPONSE_BYTES=2097152
PRODUCT_TO_MCP_ALLOW_LEGACY_RELEASES=false
PRODUCT_TO_MCP_MCP_BEARER_TOKEN=
PRODUCT_TO_MCP_SECRET_ENCRYPTION_KEY=long-stable-secret
```

Frontend environment field:

```
VITE_PRODUCT_TO_MCP_API_BASE_URL=https://api.your-domain.com
```

Important: `PRODUCT_TO_MCP_PUBLIC_BASE_URL` must be the backend HTTPS URL, not the frontend URL.

Step 7: Publish to Smithery

Field	Required	Notes
Smithery namespace	Yes	Example: @acme or acme
Server name/slug	Yes	Example: demo-store-mcp
Smithery API key	Yes	Submitted once, then cleared
Permission to publish/update server	Yes	Must exist in customer's Smithery account

Publish Readiness Check	Needed
Public HTTPS MCP URL	Yes
Streamable HTTP MCP endpoint	Yes
initialize works	Yes
tools/list works	Yes
Smithery API key valid	Yes

Guardrails

- The user must explicitly select capabilities exposed by an MCP release.
- A source document never grants permission by itself.
- Secrets must not appear in tool schemas, manifests, logs, browser state, or responses.
- Each new source revision creates a new release; active releases are never mutated in place.